1. **Computer network:** A group of interconnected devices (computers, phones, servers, printers) that can communicate and share data and resources using wired or wireless links and agreed rules called protocols.

2. **Client vs server:** A client requests services or data (e.g., your browser). A server provides them (e.g., a web server) and responds to many clients.

3. **LAN (Local Area Network):** A network covering a small area such as a home, school, or office, usually with high speed and under one owner's control.

4. **WAN (Wide Area Network):** A network spanning large geographic areas (cities, countries) and connecting multiple LANs. The Internet is the largest WAN.

5. **Switch:** Connects devices within a LAN and forwards data only to the intended device, using MAC addresses.

6. **Router:** Connects different networks (e.g., your LAN to the Internet) and forwards packets between them using IP addresses, choosing the best path.

7. **Firewall:** Hardware or software that monitors and filters incoming and outgoing traffic based on security rules, blocking unauthorized or malicious connections.

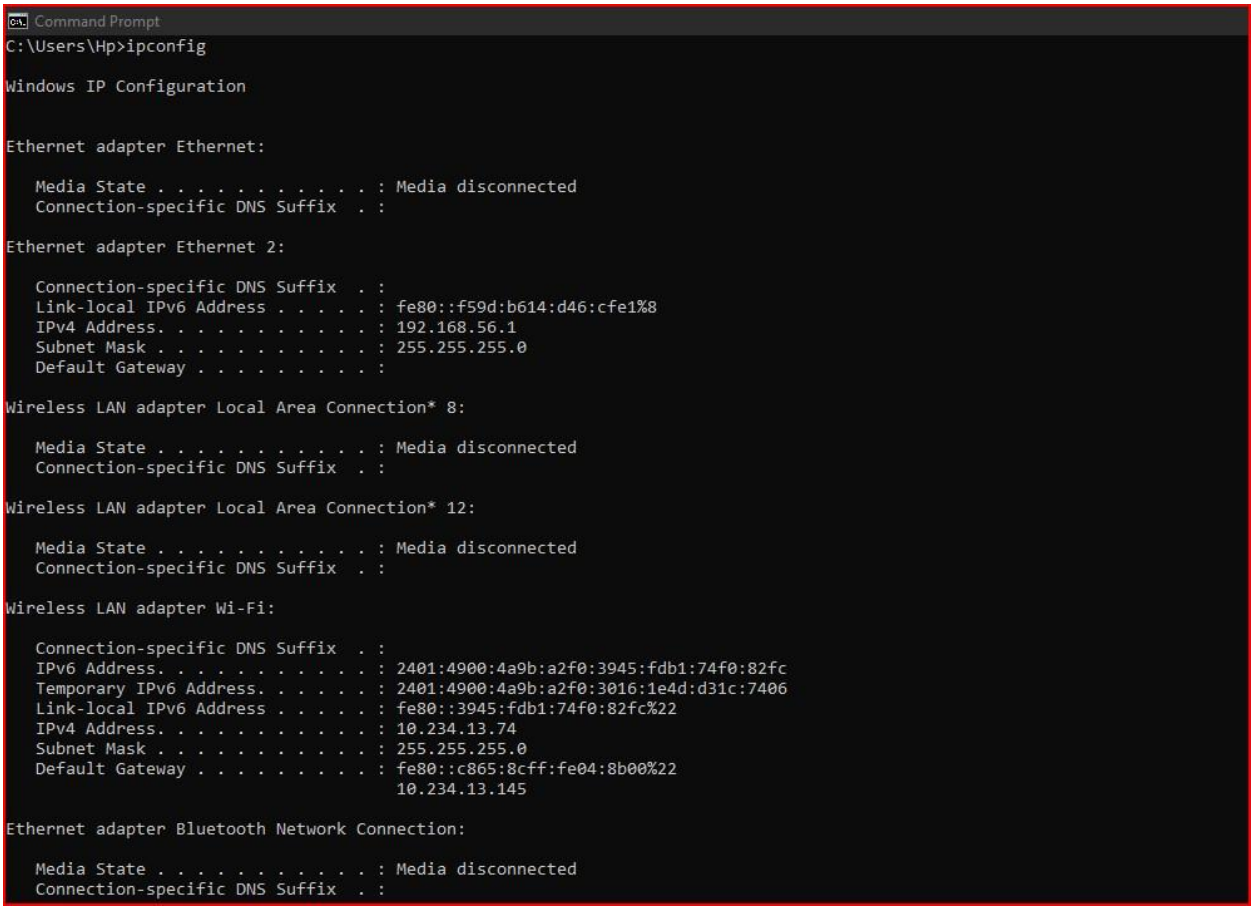
8. **IP address:** A logical numerical address assigned to a device on a network (e.g., 192.168.1.10) that identifies it and lets it send and receive data. It can change depending on the network.

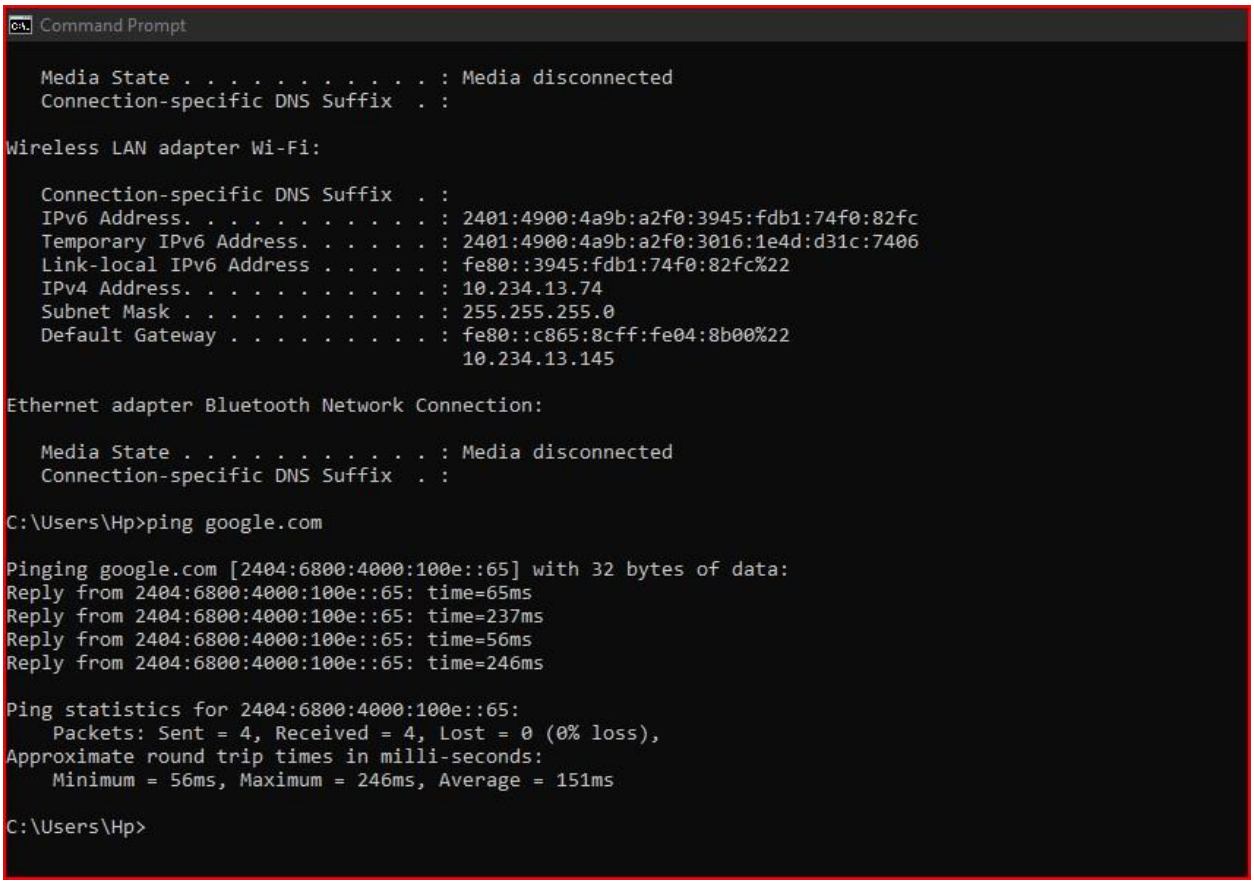
9. **MAC address:** A unique 48-bit hardware address burned into a network interface card (e.g., 00:1A:2B:3C:4D:5E), used for communication within a local network.

10. **Default gateway:** The router address a device sends traffic to when the destination is outside its own network. It is the “exit door” to other networks and the Internet.

11. **DNS (Domain Name System):** Translates human-readable domain names (e.g., www.google.com) into IP addresses so devices can locate each other.

12. **Why networking matters for cybersecurity:** Attacks, defenses, and evidence all travel over networks. Understanding protocols, addressing, and traffic flow lets you spot suspicious activity, configure firewalls and segmentation, analyze packets, understand attacks (like spoofing, sniffing, DNS poisoning, DoS), and secure systems properly.





```
Command Prompt

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2401:4900:4a9b:a2f0:3945:fdb1:74f0:82fc
   Temporary IPv6 Address. . . . . . : 2401:4900:4a9b:a2f0:3016:1e4d:d31c:7406
   Link-local IPv6 Address . . . . . : fe80::3945:fdb1:74f0:82fc%22
   IPv4 Address. . . . . . . . . . . : 10.234.13.74
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::c865:8cff:fe04:8b00%22
                                       10.234.13.145

Ethernet adapter Bluetooth Network Connection:

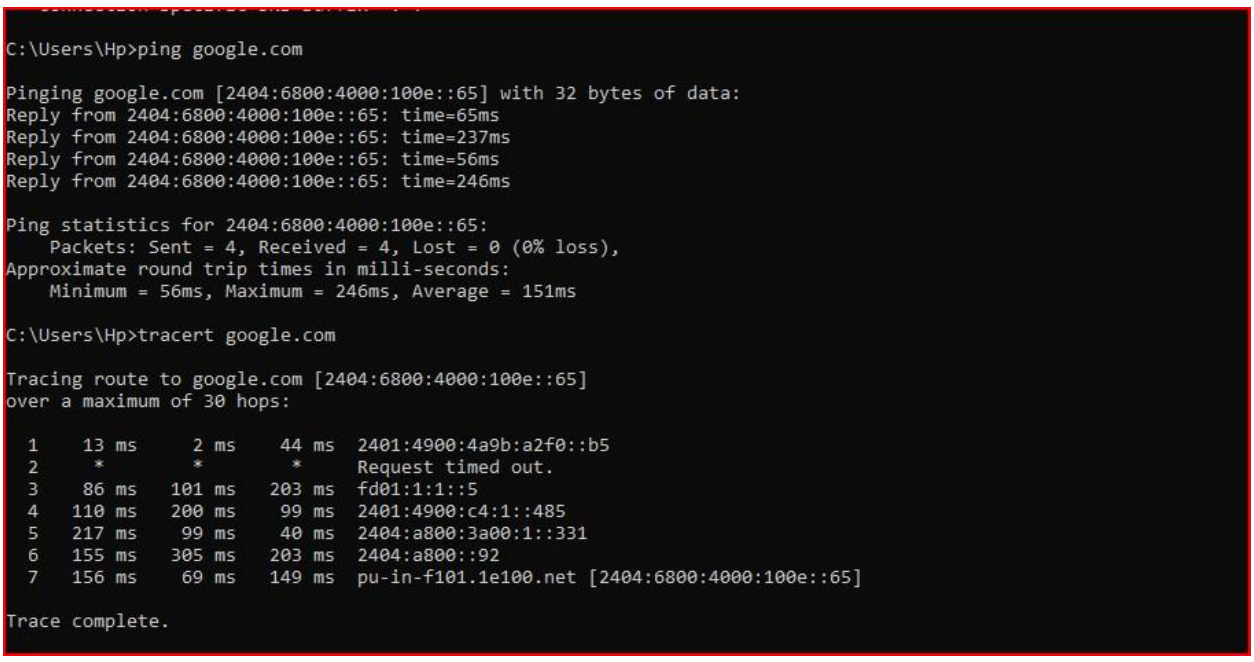
   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

C:\Users\Hp>ping google.com

Pinging google.com [2404:6800:4000:100e::65] with 32 bytes of data:
Reply from 2404:6800:4000:100e::65: time=65ms
Reply from 2404:6800:4000:100e::65: time=237ms
Reply from 2404:6800:4000:100e::65: time=56ms
Reply from 2404:6800:4000:100e::65: time=246ms

Ping statistics for 2404:6800:4000:100e::65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 56ms, Maximum = 246ms, Average = 151ms

C:\Users\Hp>
```



```
C:\Users\Hp>ping google.com

Pinging google.com [2404:6800:4000:100e::65] with 32 bytes of data:
Reply from 2404:6800:4000:100e::65: time=65ms
Reply from 2404:6800:4000:100e::65: time=237ms
Reply from 2404:6800:4000:100e::65: time=56ms
Reply from 2404:6800:4000:100e::65: time=246ms

Ping statistics for 2404:6800:4000:100e::65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 56ms, Maximum = 246ms, Average = 151ms

C:\Users\Hp>tracert google.com

Tracing route to google.com [2404:6800:4000:100e::65]
over a maximum of 30 hops:

  1    13 ms     2 ms    44 ms  2401:4900:4a9b:a2f0::b5
  2     *        *        *     Request timed out.
  3    86 ms   101 ms   203 ms  fd01:1:1::5
  4   110 ms   200 ms    99 ms  2401:4900:c4:1::485
  5   217 ms    99 ms    40 ms  2404:a800:3a00:1::331
  6   155 ms   305 ms   203 ms  2404:a800::92
  7   156 ms    69 ms   149 ms  pu-in-f101.1e100.net [2404:6800:4000:100e::65]

Trace complete.
```